

# **CSE 251**

# **Electronic Devices and**

# **Circuits**

## **Lecture 1**



Inspiring Excellence

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**University**

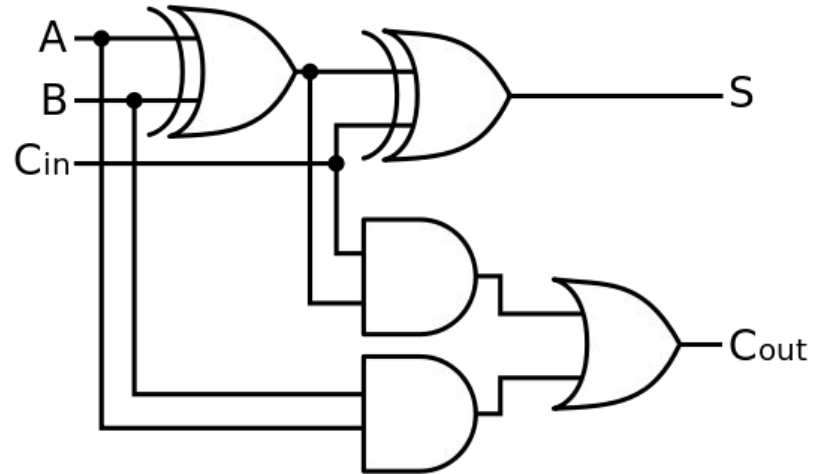
***Email: [ankan.ghosh@bracu.ac.bd](mailto:ankan.ghosh@bracu.ac.bd)***

## Mathematical Operations

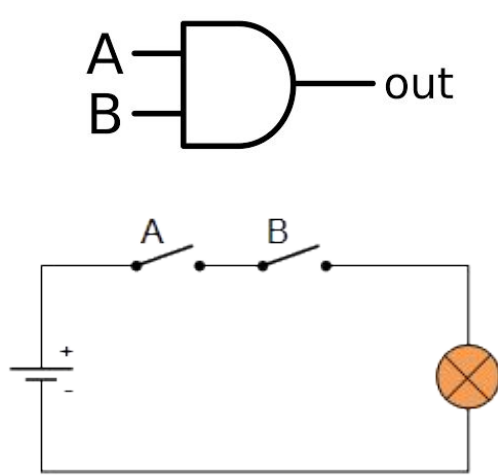
- Addition:  $4 + 5 \square 9$  ( $0100 + 0101 \square 1001$ )
- Subtraction:  $10 - 9 \square 1$  ( $1010 + 0111 \square 0001$ )
- Multiplication:  $5 \times 4 = 4 + 4 + 4 + 4 + 4 = 20$
- Division:  $10 / 2 = 2$  can be subtracted from 10, 5 times

# Digital Logic Circuit

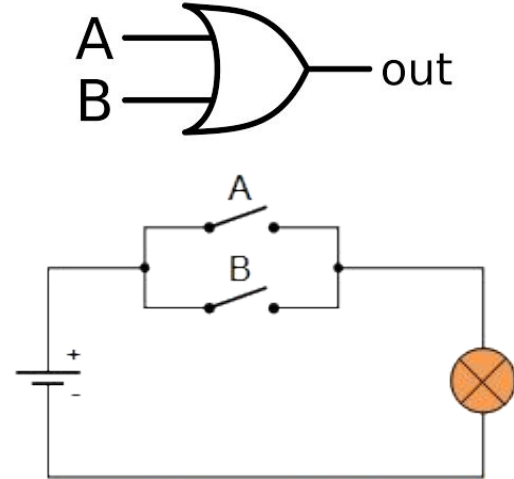
**Addition:**  $4 + 5 \square 9$  ( $0100 + 0101 \square 1001$ )



# Logic gates are basically switches



**AND Gate**



**OR Gate**

The faster you can operate these switches, the faster you can complete the functions!!!

# How can we make these switches?

- **Mechanical switch**

- Bulky and heavy
- Mechanical wear over time
- Noisy
- Ultra slow
- Requires lots of energy to operate



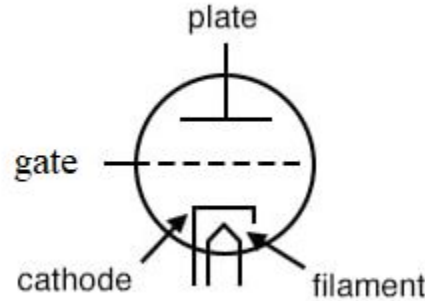
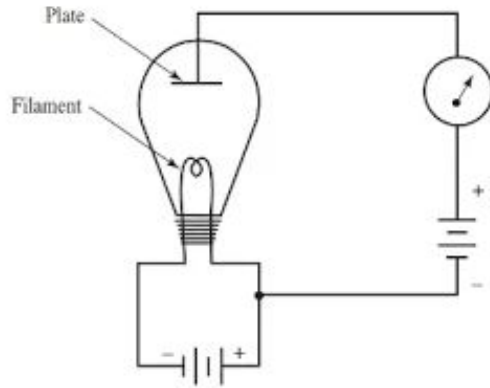
**SPST  
Switch**



**Electromechanical  
Relay**

# How can we make these switches?

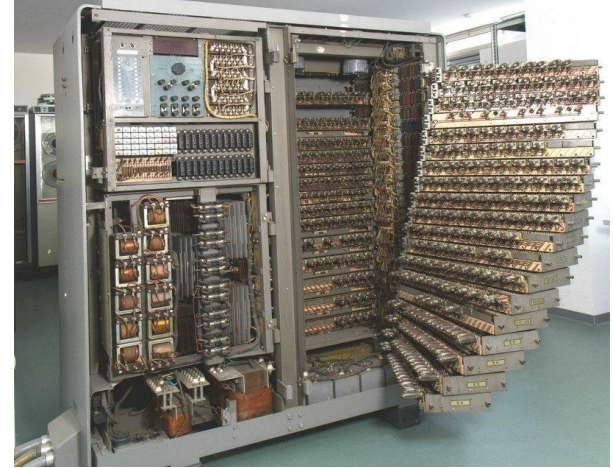
- **Vacuum tube**



- To ensure current passes along one direction and stops flowing in the other direction
- Gate allows us to control this is a more robust way

# How can we make these switches?

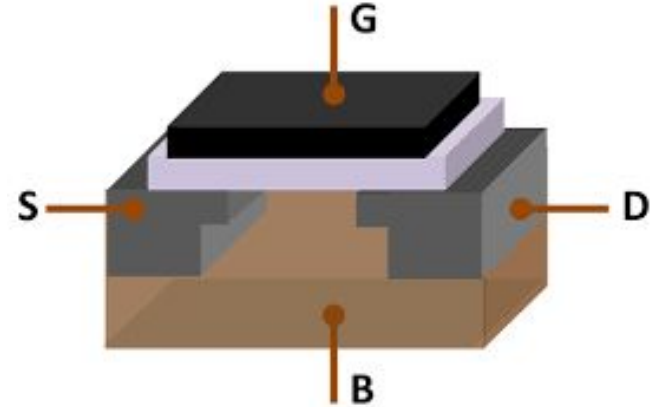
- **Vacuum tubes**
  - Bulky
  - Lots of energy
  - Not scalable



# How can we make these switches?

- **Electronic switches**

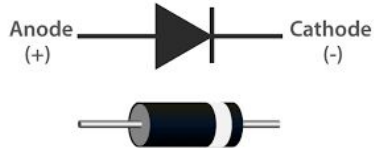
- No moving parts
- Scalable
- High speed



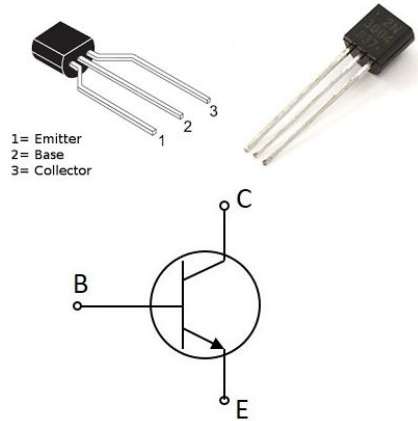
**5GHz computer □ 5 billion operations per second !!!**

# What are electronic circuits?

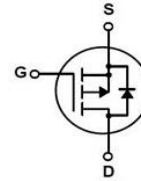
- Any circuit consisting of semiconductors.



Diode



BJT



MOSFET

- **Transistor:** probably the most impactful invention of the present world
- **Why do we need this course?**

**High Level Programming** → **Assembly language** → **Machine language** →

**Architecture** (C, C++, etc.) (x86, ARM, CUDA, etc.) (100110) (RISC, CISC, etc.)

...**Architecture** → **System level** (Mux) → **Gate** (AND, OR, etc.) → **Transistor**

Our World  
in Data

## in Data

\_\_\_\_\_

50,000,000,000



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# Why do we need electronic circuits?

## **Digital Electronics**

- Boolean logic
- Addition, subtraction, multiplication, division

## **Analog Electronics**

- Amplifiers, radio transmitter and receivers, modulator

## **Power Electronics**

- Motor control
- AC to DC conversion or vice versa
- HVDC circuits
- Charge control circuits

Thank you!